

2.1

a 9 decimal to binary

$$\begin{array}{r}
 9 \div 2 = 4,5 \quad 1 \\
 4 \div 2 = 2 \quad 0 \\
 2 \div 2 = 1 \quad 0 \\
 1
 \end{array}$$

9 decimal = 1001 binary

$$9_{10} = 1001_2$$

b 10101 binary to decimal

$$\begin{array}{ccccccc}
 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & & \\
 1 & 0 & 1 & 0 & 1 & &
 \end{array}$$

$$1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 10101$$

$$16 + 0 + 4 + 0 + 1 = 10101$$

$$\therefore 10101 (\text{binary}) = 21 (\text{decimal})$$

$$10101_2 = 21_{10}$$

c. 16 decimal to hexadecimal

$$16 \div 16 = 1 \quad R 0$$

$$1 \div 16 = 0,625 \quad R 1$$

$$16_{10} = 10_{16} \Rightarrow 1 \times 16^1 + 0 \times 16^0 = 16 + 0 = 16$$

d 3F (hexadecimal) to decimal

$$3F_{16} = x_{10}$$

$$\begin{array}{c}
 \downarrow \quad \downarrow \\
 16^1 \quad 16^0
 \end{array}$$

$$3 \times 16^1 + 15 \times 16^0$$

$$= 48 + 15$$

$$3F_{16} = 63_{10}$$

$$F = 15$$

e F hexadecimal to binary

$$F_{16} = 15_{10} = X_2$$

$$F=15$$

$$15 \div 2 = 7,5 \text{ R1}$$

$$7 \div 2 = 3,5 \text{ R1}$$

$$3 \div 2 = 1,5 \text{ R1}$$

$$1 \div 2 = 0,5 \text{ R1}$$

$$\therefore F_{16} = 1111_2$$

f $1001_2 0000_2$ to hexadecimal

$$\begin{array}{ccccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \downarrow 3 & \downarrow 2 & \downarrow 1 & \downarrow 0 & \downarrow 3 & \downarrow 2 & \downarrow 1 & \downarrow 0 \\ 8 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array}$$

$$9_{10} \quad 0_{10} = 90_{16}$$

$$\therefore 10010000 = 90_{16}$$

2,2 a) $101010 + 10101$

$$\begin{array}{r} 101010 \\ + \\ 10101 \end{array}$$

$$= 111111$$

$$= 111111_2$$

b) $111011 + 101101$

$$\begin{array}{r} 111011 \\ + \\ 101101 \end{array}$$

$$= 1101000_2$$

$$= 1101000_2$$

$$= 1101000_2$$

c) $9_{10} + 7_{10}$

$$= 9 + 7 = 16_{10}$$

d) $17 + BA_{16}$

$$A_{16} = 10_{10}$$

$$B_{16} = 11_{10}$$

d) $17 + BA$

$$A_{16} = 10$$

$$B_{16} = 11$$

$$17$$

$$7_{10} + 10_{10} = 17_{10}$$

$$= 17_{10}$$

$$11_{16} = 17_{10}$$

$$11_{10} + 1_{10} + 1_{10}$$

$$D_{16} = 13_{10}$$

$$17$$

$$AB$$

$$D1$$

$$\therefore 17 + BA = D1_{16}$$

3.1

a) ~~76543210~~ 10101010

01010101

2 bytes ~~16 bits~~

$$1 \times 2^7 + 1 \times 2^5 + 1 \times 2^3 + 1 \times 2^1 = 1 \times 2^6 + 1 \times 2^4 + 1 \times 2^2 + 1 \times 2^0$$

$$128 + 32 + 8 + 2 = 64 + 16 + 4 + 1$$

$$A = 170$$

$$B = 85$$

$$\text{byte } A \times 256 + \text{byte } B =$$

$$170 \times 256 + 85$$

$$= 43520 + 85$$

$$= 43605_{10}$$

b) $\overset{3}{F}\overset{2}{F}\overset{1}{F}\overset{0}{F} \rightarrow \text{hexadecimal}, F = 15$

$$15 \times 16^3 + 15 \times 16^2 + 15 \times 16^1 + 15 \times 16^0$$

$$= 61440 + 3840 + 240 + 16$$

$$= 65,536_{10}$$

c) $\overset{0}{9}\overset{1}{2}\overset{2}{7} \overset{0}{2}\overset{1}{3}\overset{1}{1}$

$$\text{byte } 1 \times 256 + \text{byte } 2$$

$$= 127 \times 256 + 231$$

$$= 32512 + 231$$

$$= 32743_{10}$$

d) $\overset{A}{000000001} \overset{B}{000000001}$

~~(1x20)~~

$$(1 \times 2^0) \times 256 + 1 \times 2^0$$

$$= 1 \times 256 + 1$$

$$= 257_{10}$$

3.2

a) 37

37	2	RI
18	2	RO
9	2	RI
4	2	RO
2	2	RO
1		

$$37_{10} = 100101_2$$

In 8 bits = 00100101

b) -7

7	2	RI
3	2	RI
1		

7 = 111 → in 8 bits 00000111
we flip every 0 to 1 and
the opposite 1 to 0 and add 1

$$\therefore 11111000 + 00000001$$

$$= 11111000 \\ + 00000001 \\ \hline 11111001$$

$$\Rightarrow -7 = 11111001_2$$

c) 120

120	2	RO
60	2	RO
30	2	RO
15	2	RI
7	2	RI
3	2	RI
1		

$$120 = 1111000$$

In 8 bits = 01111000

e) 37 - 7

from a and b → 11111001
↓
00100101

$$00100101 \\ + 11111001 \\ \hline$$

$$10001110$$

$$\therefore 37 - 7 = 00011110$$

d) 10

10 / 2 R 0

5 / 2 R 1

2 / 2 R 0

1

$$10_{10} = 1010$$

in 8 bits = 00001010

f) 120 + 10

120 / 2 0

60 / 2 0

30 / 2 0

15 / 2 1

7 / 2 1

3 / 2 1

1

$$1111000 + 0001010$$

$$1111000 +$$

$$0001010$$

$$10000010$$

$$120_{10} + 10_{10} = 10000010$$

3.3 a) 0.3 8 bits

0.3 x 2 R 0

0.6 x 2 R 1

0.2 x 2 R 0

0.4 x 2 R 0

0.8 x 2 R 1

0.6 x 2 R 1

0.2 x 2 R 0

0.4 x 2 R 0

0.8 x 2 R 1

$$0.010011001 - \frac{1}{512}$$

$$\frac{1}{4} + \frac{1}{32} + \frac{1}{64} + \frac{1}{512} + \frac{1}{1024} = 0.3$$

Sign = 0 (0.3 is positive)

exponent = -2

mantissa = 0011

$$\text{bit bias} = \text{exponent} + \text{bias}$$

$$-2 + 3 = -1$$

0.3 in 8 bit = 0 001 0011

b) 0.3 (16 bit)

1 Signed 5 exponent
10 mantissa

$$0011001100110011 = 1.100... \times 2^{-2}$$

15 bias

~~exponent~~ + bias = ~~8~~ Stored expo

7	2	1	= -2 + 15 = 13 ₁₀ = 1101 ₂
6	2	0	
3	2	1	
1	2	1	

0.3 = (16 bits) → 0 01101 0011001100

↓ exponent mantissa

positive signed

c) -14,731567149

8 bit

14	2	0
7	2	1
3	2	1
1	2	1
0		

$$14_{10} = 01110_2$$

0.731	x 2	1
0.46	x 2	0
0.92	x 2	1
0.84	x 2	1
0.68	x 2	1
0.36	x 2	0
0.72	x 2	1
0.44	x 2	0
0.88	x 2	1
0.76	x 2	1
0.52	x 2	1
0.04		0

$$0.731..._{10} = 101110101110$$

$$= 01110.1011101011..._2$$

$$74,731 \dots = 1111^0 1011101011 \dots_2$$

$$\text{Sign} = 1 \quad = 1.1101011101011 \dots_2 \times 2^3$$

Stored ex = exponent + bias

$$3 + 3$$

$$= 6_{10} = 110_2$$

$\therefore 1 \quad 110 \quad 1101$

d) -14,731 ... in 16 bit

From c ... stored exponent = exponent + bias

18	2	0
9	2	1
4	2	0
2	2	0
1	2	1

Sign	exponent	mantissa
$\therefore 1$	10010	1011011101

~~$$= 18_{10} = 01010_2$$~~

~~$$= 18_{10} = 10010_2$$~~

4)

binary a) $1010 = 2^3 + 2^1 = 10$

b) $0101 = 2^2 + 1 = 5$

c) $0110 = 2^2 + 2^1 = 6$

d) $1001 = 2^3 + 1 = 9$

a

b

c

d

$$\text{hexa}^a = A$$

$$b = 5$$

$$c = 6$$

$$d = 9$$

5 a) $!(True | False)$

first we start with the brackets
"|" one if one of them is true then ~~its~~
the answer is true

so $!(True) \Rightarrow$ "!" if A is true
then B is false

$\therefore a = False$

b $(True \& True) \wedge False$

"&" A and B ~~must be true~~
if they are the same "True" then
it's true

$\therefore True \wedge False$ (OR = when A and B are different
it's True)

$b = \therefore True$

c) $\sim (0110 \& 1010)$

$\sim (FTTF \& TFTF)$

$\sim (FFTF)$

$\sim (0010)$

$c = 1101$

d) $(0110 \& \sim 1010) | 0110$

$(FTTF \wedge \sim TFTF) | FTTF$

$(TTFF) | FTTF$

$= TTTF$

$d = 1110$

6

a)

A	B	=	A	B	&		^	!A	!B
1	1	0	1	1	1	1	0	0	0
1	0	0	1	0	0	1	1	0	1
0	1	0	0	1	0	1	1	1	0
0	0	1	0	0	0	0	0	1	1

$$\therefore (A | B) !$$

$$(1110) !$$

$$\Rightarrow \boxed{= (0001)}$$

b)

A	B	=	A	B	&		^	!A	!B
1	1	0	1	1	1	1	0	0	0
1	0	0 1	1	0	0	1	1	0	1
0	1	0	0	1	0	1	1	1	0
0	0	1	0	0	0	0	0	1	1

$$\therefore (B) !$$

$$= (1010) !$$

$$\boxed{b = 0101}$$